INCH-POUND

A-A-52126
May 17, 1993

COMMERCIAL ITEM DESCRIPTION

POLISHER, PORTABLE, ELECTRIC

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-P-40096.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers the requirements for one type of portable, hand-held, electric polisher with carrying case.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The polisher shall be a hand-held, electric motor-driven unit, designed to impart a rotary motion to a circular polishing bonnet mounted on a flexible pad attached to a spindle driven by the motor. The polisher shall have an electric motor, housing, bonnet, flexible mounting pad, bearings, rear handle, removable front handle, electrical components, and a carrying case. The unit shall conform to the characteristics listed in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5130

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TABLE I. Polisher characteristics

Duty	Nominal size (inches)	Pad size (diam. in inches)	Full-load power 1/ output (hp) (min.)	No-load spindle speed (RPM) (min.)	Rated current (ampere (min.)	Length without pad (inches) (max.)	Weight without pad or cable (pounds) (max.)
Inter-mittent	7	7	1/2	1700	5.0	17	13

1/ Full load power shall be determined at minimum rated current.

2.1.1 Motor. The motor shall be enclosed and mounted within the polisher housing. The motor shall be of the intermittent duty type and shall conform to NEMA MG-1 and the electrical characteristics specified in table I.

2.1.2 Power supply cable. Each polisher shall be furnished with a cable and plug assembly. At the point of cable entry into the polisher, a heavy duty strain relief shall be provided. The cable shall not be less than 8 feet long consisting of a three wire electrical cord, type S, SO, SJ, ST, SJO or SJT in accordance with NFPA Standard No. 70. The cable shall terminate in a three-way, integrally molded connection plug conforming to UL Standard No. 498 requirements rated at 15 amperes, and 120 volts.

2.1.3 Bearings. The armature or rotor shaft shall be equipped with sealed ball or roller bearings. The intermediate shaft and driving shaft shall have ball, needle, or roller bearings. Needle bearings shall be used only when the hardness of the supporting shaft, size, thrust bearings, and seals are adequate. Thrust bearings shall take the end thrust of the polisher spindle. The ball and roller bearings shall conform to all applicable standards of the Anti-Friction Bearing Manufacturer Association standards for Ball and Roller Bearings.

2.1.4 Switch. Switches shall conform to the safety requirements of UL Standard No. 45. Switches shall be of the quick-break type and shall have either an automatic release or an "On/Off" switch. Automatic release switches shall have a lock for holding the switch in the "On" position.

2.1.5 Housing. The housing shall be cast aluminum alloy, plastic, or a combination of both.

2.1.6 Handles. The rear handle shall be fastened to the rear of the housing. The front handle shall be removable and capable of being attached to either side of the housing for left-hand or right-hand use.

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2.1.7 Backing pad. The polisher shall have a 7 inch semi-flexible backing pad for the polishing bonnet, and the pad shall be provided with a threaded or tapped adaptor for attaching to the polisher drive shaft.

2.1.8 Bonnet. The polishing bonnet shall be fabricated from either lamb's wool or wool yarn sewed on to a canvas backing pad. The polishing bonnet shall be 7-1/2 inches in diameter and shall be provided with a flexible edging to fit snugly over the pad edge. Each polisher shall be provided with three 7-1/2 inch diameter polishing bonnets.

2.1.9 Carrying case. The carrying case shall be either metal or plastic, and shall have a hinged cover and a carrying handle.

2.2 Electrical requirements. The polisher shall be designed to operate on 120 volts, 60 hertz, single phase power, or as specified (see 5.2).

2.3 Finish. The finish coat of the polisher and carrying case shall be continuous, smooth, and adherent with no sags, runs, cracks, pits, rust, rough grinds, or toolmarks showing.

2.4 Marking for identification. Identification shall be placed on the housing showing the current manufacturer's name, model number, serial number, maximum size or capacity of the polisher, voltage, current rating (amperes) and revolutions per minute (RPM).

2.5 Workmanship. The polisher shall be free from defects such as: fractures, splits, punctures, tears, dents, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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3.3 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

4. PACKAGING

4.1 Packaging, packing, and marking. The polisher shall be packaged, packed, and marked in accordance with domestic or overseas requirements of ASTM D 3951, as applicable (see 5.2).

5. NOTES

5.1 Intended use. The intermittent duty, portable electric polisher is intended for use in polishing lacquered and painted surfaces on automobiles or furniture. The polisher can be used on flat or curved surfaces.

5.2 Acquisition requirements. Acquisition requirements must specify the following:

a. Title, number, and date of this document.
b. Electrical requirements (if different than 2.2).
c. Type of packaging, packing, and marking requirements required (see 4.1).

5.3 Sources of documents.

5.3.1 Sources of non-Government association documents.

AFBMA Standards for Ball and Roller Bearings and Balls, are available from:

The Anti-Friction Bearing Manufacturers Association (AFBMA)
1101 Connecticut Avenue, N.W., Suite 700
Washington, DC 20036

ASTM D 3951 - Standard Practice for Commercial Packaging, is available from:

The American Society for Testing and Materials (ASTM)
1916 Race Street
Philadelphia, PA 19103-1187

MG-1 - Motors and Generators, is available from:

National Electrical Manufacturers Association (NEMA)
2101 L Street, NW, Suite 300
Washington, DC 20037

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NFPA Standard No. 70 - National Electrical Code, is available from:

National Fire Protection Association (NFPA)
1 Batterymarch Park
Quincy, MA 02269

UL Standard No. 45 - Portable Electric Tools
UL Standard No. 498 - Attachment Plugs and Receptacles
These standards are available from:

Underwriters Laboratories (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.3.2 Sources of Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review activities

Army - CR, ME
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project 5130-0356

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-52126	1993 May 17

3. DOCUMENT TITLE
POLISHER, PORTABLE, ELECTRIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

(a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198/21

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